

$^{32}\text{S}(\alpha, \text{p})$  2019Se09, 1973Go16

$J^\pi=0^+$  for  $^{32}\text{S}$  ground state.

**2019Se09:** a 21-MeV  $\alpha$  beam was produced from the 10-MV FN tandem Van de Graaff accelerator at Triangle Universities Nuclear Laboratory (TUNL). Targets were  $15.9\text{-}\mu\text{g}/\text{cm}^2$   $^{\text{nat}}\text{S}$  in cadmium sulfide and  $46.3\text{-}\mu\text{g}/\text{cm}^2$   $^{\text{nat}}\text{S}$  in antimony sulfide. Reaction products were momentum-analyzed by the TUNL high-resolution Enge split-pole magnetic spectrograph and detected using a high-resolution position-sensitive focal-plane detector consisting of two position-sensitive gas avalanche counters: a gas proportionality energy loss section and a residual energy scintillator (FWHM=24 keV). Measured  $\sigma(E_p, \theta)$ , at  $\theta=10^\circ$  to  $50^\circ$ . Deduced levels,  $J$ ,  $\pi$  from one-step finite range triton transfer FRESKO-DWBA analysis. Deduced the thermonuclear  $^{34}\text{S}(\text{p}, \gamma)^{35}\text{Cl}$  reaction rate via Monte Carlo methods.

**1973Go16:** 19.991-, 20.025- and 20.284-MeV  $\alpha$  beams were produced from the University of Notre Dame FN tandem Van de Graaff accelerator. Natural sulfur targets were prepared by vacuum evaporation of CdS onto  $20\text{-}\mu\text{g}/\text{cm}^2$  carbon backings. Protons were analyzed with broad-range magnetic spectrographs at  $15^\circ$ ,  $30^\circ$ ,  $70^\circ$ ,  $90^\circ$ , and  $120^\circ$  and detected using nuclear track plates at the focal planes. Measured  $\sigma(E_p, \theta)$ . Deduced 102 levels.

**1971Au07:** a 28.8-MeV  $\alpha$  beam of 0.1-0.5  $\mu\text{A}$  was produced from the Naval Research Laboratory (NRL) isochronous cyclotron. The target was made by evaporating natural PbS onto a  $30\text{-}\mu\text{g}/\text{cm}^2$  carbon backing, and the effective  $^{32}\text{S}$  thickness was  $163\text{ }\mu\text{g}/\text{cm}^2$ . Protons were detected using two counter telescopes with FWHM=250 keV. Measured  $\sigma(E_p, \theta)$ . Deduced levels,  $J$ ,  $\pi$ , L-transfers, and J-dependence for the g.s. and 1763-keV level from local zero-range DWUCK-DWBA analysis of the measured  $\sigma(\theta)$ .

Other: **1996So21** (cross sections).

 $^{35}\text{Cl}$  Levels

| <u>E(level)<sup>†</sup></u> | <u><math>J^\pi</math> &amp;</u> | <u>L<sup>a</sup></u> | <u>Comments</u>                                                                                       |
|-----------------------------|---------------------------------|----------------------|-------------------------------------------------------------------------------------------------------|
| 0                           |                                 | 2                    | $J^\pi$ : $3/2^+$ assumed in <b>1971Au07</b> for investigation of J-dependence of L=2 transfer.       |
| 1217.4 26                   |                                 |                      |                                                                                                       |
| 1760.1 24                   |                                 | (2)                  | $J^\pi$ : $5/2^+$ assumed in <b>1971Au07</b> for investigation of J-dependence of L=2 transfer.       |
|                             |                                 |                      | L: The parentheses are added by evaluators based on the DWBA fit in <b>1971Au07</b> .                 |
| 2644.9 23                   |                                 |                      |                                                                                                       |
| 2693?                       |                                 |                      |                                                                                                       |
| 3001.5 31                   |                                 |                      |                                                                                                       |
| 3158.2 25                   |                                 |                      |                                                                                                       |
| 3945.2 23                   |                                 |                      | E(level): weighted average of 3943.7 23 ( <b>1973Go16</b> ) and 3947.9 31 ( <b>2019Se09</b> ).        |
| 4056.9 27                   | ( $3/2^-$ ) <sup>b</sup>        |                      | E(level): 4059.12 quoted from <b>2011Ch48</b> and used for internal calibrations by <b>2019Se09</b> . |
| 4111.1 29                   | ( $7/2^+$ ) <sup>b</sup>        |                      | E(level): weighted average of 4110.2 29 ( <b>1973Go16</b> ) and 4112.0 30 ( <b>2019Se09</b> ).        |
| 4177.9 24                   | ( $3/2^-$ ) <sup>b</sup>        |                      | E(level): weighted average of 4177.5 24 ( <b>1973Go16</b> ) and 4179 4 ( <b>2019Se09</b> ).           |
| 4346.6 24                   |                                 |                      | E(level): 4347.82 quoted from <b>2011Ch48</b> and used for internal calibrations by <b>2019Se09</b> . |
| 4625.3 28                   |                                 |                      | E(level): weighted average of 4624.4 31 ( <b>1973Go16</b> ) and 4626.1 28 ( <b>2019Se09</b> ).        |
| 4770.4 26                   |                                 |                      | E(level): weighted average of 4770.8 26 ( <b>1973Go16</b> ) and 4769.9 28 ( <b>2019Se09</b> ).        |
| 4859.2 <sup>‡</sup> 28      |                                 |                      |                                                                                                       |
| 4881.8 29                   |                                 |                      | E(level): weighted average of 4883.1 29 ( <b>1973Go16</b> ) and 4878 5 ( <b>2019Se09</b> ).           |
| 5006.9 <sup>‡</sup> 27      |                                 |                      |                                                                                                       |
| 5158.9 29                   | ( $5/2^+$ ) <sup>b</sup>        |                      | E(level): weighted average of 5161.7 33 ( <b>1973Go16</b> ) and 5156.8 29 ( <b>2019Se09</b> ).        |
| 5209.5 28                   | ( $5/2^+$ )                     |                      | E(level): weighted average of 5206.6 37 ( <b>1973Go16</b> ) and 5211.1 28 ( <b>2019Se09</b> ).        |
| 5402.0 29                   |                                 |                      |                                                                                                       |
| 5407.2                      | ( $11/2^-$ ) <sup>b</sup>       |                      | E(level): quoted from <b>2011Ch48</b> and used for internal calibrations by <b>2019Se09</b> .         |
| 5531 <sup>‡</sup> 5         |                                 |                      |                                                                                                       |
| 5586.0 26                   |                                 |                      | E(level): 5576 tentative level from <b>1973Go16</b> .                                                 |
| 5594.6 26                   |                                 |                      | E(level): weighted average of 5591.8 32 ( <b>1973Go16</b> ) and 5596.5 26 ( <b>2019Se09</b> ).        |
| 5633.6 30                   |                                 |                      | E(level): weighted average of 5633.1 32 ( <b>1973Go16</b> ) and 5634 3 ( <b>2019Se09</b> ).           |
| 5645 <sup>‡</sup> 4         |                                 |                      |                                                                                                       |
| 5653 <sup>‡</sup> 4         |                                 |                      |                                                                                                       |
| 5679.7 34                   |                                 |                      | E(level): weighted average of 5677.7 34 ( <b>1973Go16</b> ) and 5684 5 ( <b>2019Se09</b> ).           |
| 5731.2 <sup>‡</sup> 25      |                                 |                      |                                                                                                       |
| 5757 <sup>‡</sup> 4         |                                 |                      |                                                                                                       |

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$^{32}\text{S}(\alpha, \text{p})$  **2019Se09, 1973Go16 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup>  | $J^{\pi}$ &                       | Comments                                                                              |
|------------------------|-----------------------------------|---------------------------------------------------------------------------------------|
| 5808.1 31              |                                   | E(level): weighted average of 5809.2 34 (1973Go16) and 5807.1 31 (2019Se09).          |
| 5829 <sup>‡</sup> 4    | (5/2 <sup>-</sup> ) <sup>b</sup>  | E(level): 5823 tentative level from 1973Go16.                                         |
| 5928 <sup>‡</sup> 3    |                                   | E(level): other: 5927.4 35 (1973Go16).                                                |
| 6084.2 29              | (13/2 <sup>-</sup> ) <sup>b</sup> | E(level): 6087.4 quoted from 2011Ch48 and used for internal calibrations by 2019Se09. |
| 6104 <sup>‡</sup> 4    |                                   |                                                                                       |
| 6141 4                 |                                   | E(level): weighted average of 6140.2 40 (1973Go16) and 6142 4 (2019Se09).             |
| 6180 <sup>‡</sup> 3    |                                   |                                                                                       |
| 6224.4 30              |                                   | E(level): weighted average of 6224.9 36 (1973Go16) and 6224 3 (2019Se09).             |
| 6283 <sup>‡</sup> 3    | (5/2 <sup>+</sup> )               |                                                                                       |
| 6379.3 25              | (9/2 <sup>-</sup> )               | E(level): from 2019Se09. Other: 6379.0 34 (1973Go16).                                 |
| 6401.2 22              | (1/2 <sup>-</sup> )               | E(level): weighted average of 6402.4 41 (1973Go16) and 6400.9 22 (2019Se09).          |
| 6429 <sup>‡</sup> 3    | (1/2 <sup>+</sup> )               | E(level): 6427 tentative level from 1973Go16.                                         |
| 6475 <sup>‡</sup> 4    |                                   |                                                                                       |
| 6492 <sup>‡</sup> 3    | (3/2 <sup>+</sup> )               | E(level): other: 6491.9 34 (1973Go16).                                                |
| 6549 <sup>‡</sup> 3    | (1/2 <sup>+</sup> )               |                                                                                       |
| 6659.1 30              | (7/2 <sup>+</sup> )               | E(level): weighted average of 6656.0 31 (1973Go16) and 6662 3 (2019Se09).             |
| 6679.4 31              | (1/2 <sup>+</sup> )               | E(level): weighted average of 6680.8 31 (1973Go16) and 6677 4 (2019Se09).             |
| 6781.3 30              | (3/2 <sup>-</sup> )               | E(level): weighted average of 6782.8 32 (1973Go16) and 6780 3 (2019Se09).             |
| 6800 4                 |                                   | E(level): weighted average of 6802.1 42 (1973Go16) and 6795 6 (2019Se09).             |
| 6842 <sup>‡</sup> 4    | (3/2 <sup>+</sup> )               |                                                                                       |
| 6863 <sup>‡</sup> 3    | (9/2 <sup>+</sup> )               | E(level): 6867 tentative level from 1973Go16.                                         |
| 6891.6 30              | (9/2 <sup>+</sup> )               | E(level): weighted average of 6893.5 32 (1973Go16) and 6890 3 (2019Se09).             |
| 6948.6 34              | (5/2 <sup>+</sup> )               | E(level): weighted average of 6947.5 34 (1973Go16) and 6950 4 (2019Se09).             |
| 6986.3 34              | (9/2 <sup>+</sup> )               | E(level): weighted average of 6985.7 34 (1973Go16) and 6987 4 (2019Se09).             |
| 7066.2                 | (5/2 <sup>+</sup> )               | E(level): 7066.2 quoted from 2011Ch48 and used for internal calibrations by 2019Se09. |
| 7105?                  |                                   |                                                                                       |
| 7121.7 <sup>#</sup> 24 | (5/2 <sup>-</sup> )               | E(level): weighted average of 7120.8 34 (1973Go16) and 7122.1 24 (2019Se09).          |
| 7180 <sup>‡</sup> 4    | (7/2 <sup>+</sup> )               | E(level): other: 7179.0 43 (1973Go16).                                                |
| 7213.7 25              |                                   | E(level): weighted average of 7210.1 39 (1973Go16) and 7215.2 25 (2019Se09).          |
| 7230.9 30              | (3/2 <sup>+</sup> )               | E(level): weighted average of 7229.5 34 (1973Go16) and 7232 3 (2019Se09).             |
| 7301?                  |                                   |                                                                                       |
| 7348 <sup>‡</sup> 3    | (7/2)                             | E(level): other: 7348.3 44 (1973Go16).                                                |
| 7362 <sup>‡</sup> 3    |                                   |                                                                                       |
| 7413.8 30              |                                   | E(level): weighted average of 7417.5 44 (1973Go16) and 7412 3 (2019Se09).             |
| 7447 <sup>‡</sup> 3    |                                   |                                                                                       |
| 7503 4                 |                                   | E(level): weighted average of 7503.0 37 (1973Go16) and 7502 5 (2019Se09).             |
| 7566.4 35              |                                   | E(level): weighted average of 7567.6 35 (1973Go16) and 7564 5 (2019Se09).             |
| 7590 4                 |                                   | E(level): weighted average of 7587.2 40 (1973Go16) and 7595 6 (2019Se09).             |
| 7648.1 30              |                                   | E(level): weighted average of 7650.1 40 (1973Go16) and 7647 3 (2019Se09).             |
| 7674.6 <sup>‡</sup> 25 |                                   | E(level): 7671 tentative level from 1973Go16.                                         |
| 7709.9 <sup>‡</sup> 25 |                                   |                                                                                       |
| 7730?                  |                                   |                                                                                       |
| 7744.1 <sup>#</sup> 30 |                                   | E(level): weighted average of 7742.7 38 (1973Go16) and 7745 3 (2019Se09).             |
| 7770.3 33              |                                   | E(level): weighted average of 7776.3 45 (1973Go16) and 7768.5 25 (2019Se09).          |
| 7799.8 30              |                                   | E(level): weighted average of 7800.8 35 (1973Go16) and 7799 3 (2019Se09).             |
| 7868.5 35              |                                   | E(level): from 1973Go16. Other: 7867 7 (2019Se09).                                    |
| 7901 4                 |                                   | E(level): weighted average of 7900.6 38 (1973Go16) and 7899 12 (2019Se09).            |
| 7978.6 35              |                                   |                                                                                       |
| 8000.7 41              |                                   |                                                                                       |
| 8019.2 35              |                                   |                                                                                       |
| 8075.1 46              |                                   |                                                                                       |

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$^{32}\text{S}(\alpha, \text{p})$  **2019Se09,1973Go16** (continued) $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup>  | E(level) <sup>†</sup>  | E(level) <sup>†</sup>   | E(level) <sup>†</sup>   |
|------------------------|------------------------|-------------------------|-------------------------|
| 8103.8 <sup>#</sup> 42 | 8654.1 41              | 9334.0 46               | 10117?                  |
| 8121?                  | 8698?                  | 9376.0 41               | 10162.7 53              |
| 8171.7 42              | 8720.9 <sup>#</sup> 37 | 9475.5 43               | 10179.1 53              |
| 8210.2 46              | 8786.3 38              | 9508.1 <sup>#</sup> 41  | 10218.0 46              |
| 8246.0 42              | 8840.3 38              | 9525?                   | 10395.2 54              |
| 8270.6 <sup>#</sup> 42 | 8887.5 41              | 9711.8 44               | 10443?                  |
| 8292?                  | 8957.5 44              | 9740.4 40               | 10463.2 <sup>#</sup> 49 |
| 8321.0 38              | 8996.9 40              | 9799.2 47               | 10517.1 54              |
| 8398.8 38              | 9027.4 38              | 9836.1 44               | 10548.3 54              |
| 8429.2 40              | 9085.7 38              | 9870?                   | 10579.5 54              |
| 8482.3 38              | 9109.3 49              | 9968.6 <sup>@</sup> 52  | 10643.0 50              |
| 8505.1 47              | 9160.0 42              | 9992?                   | 10717?                  |
| 8532.4 43              | 9194.0 39              | 10009?                  | 10732.1 <sup>#</sup> 50 |
| 8567.9 37              | 9264.9 39              | 10075.2 <sup>@</sup> 48 | 10759.1 55              |
| 8616.9 37              | 9316.2 39              | 10098?                  |                         |

<sup>†</sup> From **1973Go16**, unless otherwise noted.

<sup>‡</sup> From **2019Se09**.

<sup>#</sup> Doublet in **1973Go16**.

<sup>@</sup> Triplet in **1973Go16**.

<sup>&</sup> From **2019Se09**, based on FRESKO-DWBA analysis of measured  $\sigma(\theta)$ , unless otherwise noted.

<sup>a</sup> From **1971Au07**.

<sup>b</sup> Original firm assignments in **2019Se09** are placed in parentheses by the evaluators when DWBA fits to the measured  $\sigma(\theta)$  were poor or when only a few data points were available for fitting.